# translationBridge MQTT/TLS/JSON

## Interface Design Description

Service ID: *"translationBridge"*

**Abstract**
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `translationBridge` service of the TranslationManager system.

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# 1 Overview

This document describes the `translationBridge` service interface of the TranslationManager system over MQTT, which provides interface and data-model translation for a consumer: finding providers that become usable through a translation bridge, building the bridge and aborting it.

It's implemented using protocol, encoding as stated in below table:

| Profile type | Type | Version |
|---|---|---|
| Transfer protocol | MQTT | 3.1.1 [1] |
| Data encryption | TLS | 1.3 [2], 1.2 [3] |
| Encoding | JSON | RFC 8259 [4] |
| Compression | N/A | - |
| Semantics | N/A | - |
| Ontology | N/A | - |

Table 1: Communication and semantics details used for the `translationBridge` service interface

This document provides the Interface Design Description IDD to the *translationBridge – Service Description* document [5]. The operations, their semantics and the data models are the same as in the HTTP/TLS/JSON IDD [6]; this document specifies the MQTT binding.

The MQTT interface is offered only when the system runs with `mqtt.api.enabled=true`; it is then registered with the interface template `generic_mqtts` (`generic_mqtt` without TLS), the broker address and port (`mqtt.broker.address`, `mqtt.broker.port`) and the base topic `arrowhead/translation/bridge/`. The standalone deployment sets `mqtt.api.enabled=false`, so the MQTT interfaces are not registered there.

# 2 Service Operations

The provider subscribes to one topic per operation, the base topic followed by the operation name:

| Operation | Request topic |
|---|---|
| `discovery` | `arrowhead/translation/bridge/discovery` |
| `negotiation` | `arrowhead/translation/bridge/negotiation` |
| `abort` | `arrowhead/translation/bridge/abort` |

A consumer publishes a request envelope (MqttRequestTemplate) to the topic and receives the answer (MqttResponseTemplate) on its own `responseTopic`. The `authentication` field carries the requester's identity according to the provider's authentication policy: `IDENTITY-TOKEN//<token>` (outsourced or internal policy), `SYSTEM//<name>` (declared policy) or the requester's X.509 certificate (certificate policy). The `status` of the response uses the HTTP status code numbers (200, 201, 204, 400, 401, 403, 404, 408, 423, 500, 503); on error the payload is an ErrorMessage.

```
{
  "traceId": "a1b2c3",
  "authentication": "IDENTITY-TOKEN//<token>",
  "responseTopic": "arrowhead/response/TemperatureDisplay/a1b2c3",
  "qosRequirement": 1,
  "params": {},
  "payload": {}
}
```

Listing 1: Request envelope.

```
{
  "status": 200,
  "traceId": "a1b2c3",
  "receiver": "TemperatureDisplay",
  "payload": {}
}
```

Listing 2: Response envelope.

## 2.1 PUBLISH `arrowhead/translation/bridge/discovery`

**Operation:** discovery
**Input:** TranslationDiscoveryRequest
**Output:** TranslationDiscoveryResponse

Returns the candidates usable by the requester through translation, with a bridge identifier.

```
{
  "candidates": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
        "addresses": [
          {}
        ],
        "device": {
```

```
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ],
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "operation": "FILTER",
  "interfaceTemplateNames": [
    "string"
  ],
  "inputDataModelId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "outputDataModelId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```


Listing 3: Payload of an discovery request.

## 2.2 PUBLISH `arrowhead/translation/bridge/negotiation`

**Operation:** negotiation
**Input:** TranslationNegotiationRequest
**Output:** TranslationNegotiationResponse

Builds the bridge to the chosen candidate and returns the bridge interface and token limits.


```
{
```

```
  "bridgeId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "target": {
    "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
    "provider": {
      "name": "TemperatureSensor",
      "metadata": {
        "location": "hall-1"
      },
      "version": "1.0.0",
      "addresses": [
        {
          "type": "IPV4",
          "address": "192.168.0.10"
        }
      ],
      "device": {
        "name": "Raspberry1",
        "metadata": {
          "location": "hall-1"
        },
        "addresses": [
          {}
        ],
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    },
    "serviceDefinition": {
      "name": "temperatureReading",
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    },
    "version": "1.0.0",
    "expiresAt": "2026-10-22T12:00:00Z",
    "metadata": {
      "unit": "celsius"
    },
    "interfaces": [
      {
        "templateName": "generic_https",
        "protocol": "https",
        "policy": "CERT_AUTH",
        "properties": {
          "accessAddresses": [
            "192.168.0.10"
          ],
          "accessPort": 8080,
          "basePath": "/temperature",
          "operations": {
            "read": {
              "method": "GET",
              "path": "/read"
            }
          }
        }
      }
    ],
    "createdAt": "2026-10-22T12:00:00Z",
    "updatedAt": "2026-10-22T12:00:00Z"
  },
  "operation": "FILTER",
  "interfaceTemplateName": "string",
  "inputDataModelId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "outputDataModelId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```


Listing 4: Payload of an negotiation request.

### 2.3 PUBLISH `arrowhead/translation/bridge/abort`

**Operation:** abort
**Input:** String

Aborts a bridge of the requester. Returns 200 if aborted, 204 if there was nothing to abort. The bridge identifier fills the `{bridgeId}` placeholder of the registered path.

```
1 "3fa85f64-5717-4562-b3fc-2c963f66afa6"
```

Listing 5: Payload of an abort request.

# 3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Each *struct* denotes a JSON Object with the named fields; fields that are not needed may be omitted.

## 3.1 struct MqttRequestTemplate

The request envelope.

| Object Field | Value Type | Description |
|---|---|---|
| "traceId" | String | Identifier chosen by the requester, returned in the response. |
| "authentication" | String | Identity of the requester (see Section 2). |
| "responseTopic" | String | Topic on which the response is published. |
| "qosRequirement" | Number | MQTT QoS of the response (0, 1 or 2). |
| "params" | Object<String> | Optional parameters (the query parameters of the HTTP binding). |
| "payload" | Object | The operation input. |

## 3.2 struct MqttResponseTemplate

The response envelope.

| Object Field | Value Type | Description |
|---|---|---|
| "status" | Number | Status code. |
| "traceId" | String | The trace identifier of the request. |
| "receiver" | String | Name of the requester. |
| "payload" | Object | The operation output, or an ErrorMessage. |

## 3.3 struct ErrorMessage

| Object Field | Value Type | Description |
|---|---|---|
| "errorMessage" | String | Error text, if the event is an error. |
| "errorCode" | Number | |
| "exceptionType" | ExceptionType | |
| "origin" | String | |

## 3.4 struct Address

A network address with its type.

| Object Field | Value Type | Description |
|---|---|---|
| "type" | String | Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`. |
| "address" | String | One network address (host name, IPv4, IPv6 or MAC address). |

## 3.5 struct DeviceResponse

A registered device.

| Object Field | Value Type | Description |
|---|---|---|
| "name" | String | Device name. |
| "metadata" | Object | Free-form metadata (JSON object). |
| "addresses" | Array<Address> | Network addresses with their types. |
| "createdAt" | DateTime | UTC time of creation. |
| "updatedAt" | DateTime | UTC time of the last change. |

## 3.6 struct ServiceDefinitionResponse

A service definition.

| Object Field | Value Type | Description |
|---|---|---|
| "name" | String | Service definition name. |
| "createdAt" | DateTime | UTC time of creation. |
| "updatedAt" | DateTime | UTC time of the last change. |

## 3.7 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

| Object Field | Value Type | Description |
|---|---|---|
| "templateName" | String | Name of the interface template, snake_case (e.g. `generic_https`). |
| "protocol" | String | Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`. |
| "policy" | String | Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies. |
| "properties" | Object | Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`. |

## 3.8 struct ServiceInstanceResponse

A registered service instance.

| Object Field | Value Type | Description |
|---|---|---|
| "instanceId" | String | Service instance identifier `<System>\|<serviceDefinition>\|<version>`, or policy instance identifier. |
| "provider" | SystemResponse | The provider system. |
| "serviceDefinition" | ServiceDefinitionResponse | The service definition. |
| "version" | Version | Version of the service instance. |
| "expiresAt" | DateTime | UTC expiry time; no value means no expiry. |
| "metadata" | Object | Free-form metadata (JSON object). |
| "interfaces" | Array<ServiceInstanceInterface Response> | Service interfaces offered by the instance. |
| "createdAt" | DateTime | UTC time of creation. |
| "updatedAt" | DateTime | UTC time of the last change. |

## 3.9 struct SystemResponse

A registered system.

| Object Field | Value Type | Description |
|---|---|---|
| "name" | String | System name. |
| "metadata" | Object | Free-form metadata (JSON object). |
| "version" | Version | Version of the system. |
| "addresses" | Array<Address> | Network addresses with their types. |
| "device" | DeviceResponse | The device hosting the system, if registered. |
| "createdAt" | DateTime | UTC time of creation. |
| "updatedAt" | DateTime | UTC time of the last change. |

## 3.10 struct TranslationBridgeCandidate

A candidate reachable through translation.

| Object Field | Value Type | Description |
|---|---|---|
| "serviceInstanceId" | String | Service instance identifier `<System>\|<serviceDefinition>\|<version>`. |
| "interfaceTemplate Name" | String | Interface template of the candidate the bridge would translate to. |

## 3.11 struct TranslationDiscoveryRequest

Discovery of translation bridges for the requester.

| Object Field | Value Type | Description |
|---|---|---|
| "candidates" | Array<ServiceInstanceResponse> | Candidate service instances (from service discovery). |
| "operation" | String | Operation the consumer wants to call. |
| "interfaceTemplate Names" | Array<String> | Interface templates the consumer can use. |
| "inputDataModelId" | String | Data model of the operation input expected by the consumer. |
| "outputDataModelId" | String | Data model of the operation output expected by the consumer. |

## 3.12 struct TranslationDiscoveryResponse

Candidates reachable through a translation bridge.

| Object Field | Value Type | Description |
|---|---|---|
| "bridgeId" | String | Identifier (UUID) of the translation bridge. |
| "candidates" | Array<TranslationBridgeCandidate> | Store entries to create. |

## 3.13 struct TranslationNegotiationRequest

Builds the bridge to the chosen candidate.

| Object Field | Value Type | Description |
|---|---|---|
| "bridgeId" | String | Identifier (UUID) of the translation bridge. |
| "target" | ServiceInstanceResponse | The chosen candidate service instance. |
| "operation" | String | QoS operation: `FILTER` or `SORT`. |
| "interfaceTemplate Name" | String | Interface template the consumer will use. |
| "inputDataModelId" | String | Data model of the operation input expected by the consumer. |
| "outputDataModelId" | String | Data model of the operation output expected by the consumer. |

## 3.14 struct TranslationNegotiationResponse

The established bridge.

| Object Field | Value Type | Description |
|---|---|---|
| "bridgeId" | String | Identifier (UUID) of the translation bridge. |
| "bridgeInterface" | ServiceInstanceInterfaceResponse | Interface of the bridge the consumer calls instead of the provider. |
| "tokenExpiresAt" | DateTime | Expiry of the bridge token. |
| "tokenUsageLimit" | Number | Number of uses allowed with the bridge token. |

## 3.15 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available.

| Type | Description |
|---|---|
| Array | A JSON Array [4]; Array<A> has elements of type A. |
| DateTime | JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`. |
| ExceptionType | One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String). |
| Number | JSON Number, an integer. |
| Object | JSON Object; Object<A> has string keys and values of type A. |
| String | JSON String. |
| Version | JSON String, semantic version `major.minor.patch`. |

# 4 References

[1] A. Banks and R. Gupta, “Mqtt version 3.1.1,” OASIS, Standard, 2014.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “translationBridge – Service Description (TranslationManager),” Eclipse Arrowhead, SD 5.2.2, 2026, translationManager-v5.2.2-translationBridge-SD.pdf.

[6] ——, “translationBridge HTTP/TLS/JSON – Interface Design Description (TranslationManager),” Eclipse Arrowhead, IDD 5.2.2, 2026, translationManager-v5.2.2-translationBridge-HTTP-IDD.pdf.

# 5 Revision History

## 5.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-22 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |
| 2 | 2026-10-30 | 5.2.2 | Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated; missing primitive definitions added. | Claude Opus 5.5 (draft) |

## 5.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.2.2 | |